

19 Ans: D

Given: $\lambda = 600 \text{ nm}$. Deduce grating spacing $d = 1 / (3.0 \times 10^5) \text{ m}$

Total number of images $= 2 n_{\text{highest}} + 1$ (1)

To det highest order of diffraction, n_{highest} :

From $d \sin \theta = n \lambda$, $\Rightarrow n_{\text{highest}} = d \sin 90 / \lambda = 5.6$

Since n must be an integer, $n_{\text{highest}} = 5$ (the largest integer which is smaller than 5.6)

Thus in (1): total number of images $= (2 \times 5) + 1 = 11$ (D)

20 Ans: B